

Security Model Design Document



End-to-End Salesforce CRM Transformation for Customer Operations and Service Management

5.1 Role Hierarchy Design

The role hierarchy for GreenGrid Utilities was developed based on the stakeholder analysis conducted during Milestone 1. The primary purpose of this hierarchy is to establish controlled data visibility across the organization while maintaining an effective reporting structure. When Organization-Wide Defaults (OWD) are configured with restrictive access settings, the role hierarchy enables records to be shared upward automatically, allowing management personnel to monitor operational activities without requiring manual sharing configurations.

To maintain consistency throughout the project, the Salesforce role structure follows the organizational groups identified in the business process analysis and project documentation.

Executive Management (Top-Level Role)

Executive Management occupies the highest position within the hierarchy and requires comprehensive visibility into customer service operations, utility asset activities, and field service performance. This access level enables management personnel to monitor organizational performance through dashboards, reports, and operational metrics.

Customer Support Team (Operational Role)

Users assigned to the Customer Support Team are responsible for handling customer inquiries, logging service requests, and managing Case records. This role operates at the operational level and maintains visibility over records required for customer support activities while limiting access to unrelated field service data.

Field Technicians (Operational Role)

The Field Technicians role represents employees responsible for conducting on-site inspections, maintenance, and repair activities. Users within this role primarily manage Field

Visit records and maintain ownership of service-related information while operating in the field environment.

5.2 Permission Sets Definition

To support the principle of least-privilege access, standard user profiles are configured with only the permissions necessary for basic job responsibilities. Additional capabilities are provided through Permission Sets, allowing specific users to perform specialized tasks without modifying baseline profile configurations.

Since the system utilizes Lookup Relationships between several objects, Permission Sets are used to grant additional access to users who require enhanced record management capabilities.

The security model includes two primary permission extensions.

5.2.1 Field Technician Mobile Access (Field_Technician_Mobile_Access)

This permission set is assigned to users within the Field Technicians role and supports mobile access for updating service activities directly from field locations.

Target Object	Object-Level Permission	Field-Level Access
Account	Read Only	Access limited to customer identification and service location information.
Case	Read Only	Visibility limited to service request descriptions and priority information.
Utility_Asset__c	Read Only	Read-only access to utility asset information and meter identification details.
Field_Visit__c	Read, Create, Update	Full update access to Repair_Status__c and Technician_Notes__c fields.

5.2.2 Customer Support Escalation Access (Support_Desk_Escalation_Access)

This permission set is intended for senior customer support personnel who require additional authority to manage escalated incidents and coordinate operational responses during major service disruptions.

Target Object	Object-Level Permission	Field-Level Access
Case	Read, Edit, Delete	Allows ownership reassignment and escalation management activities.
Field_Visit__c	Read, Edit	Supports technician assignment management and workload balancing.

5.3 Sharing Rules Configuration

To maintain data security and minimize unauthorized access, baseline sharing settings are configured with restrictive access controls. Sharing Rules are then applied selectively to provide additional visibility when specific business conditions are met.

5.3.1 Organization-Wide Defaults (OWD) Baseline

The following Organization-Wide Default settings are implemented to protect customer information, service records, and utility asset data.

Object Name	API Name	OWD Setting	Purpose
Account	Account	Private	Prevents unauthorized access to customer account information across departments.
Contact	Contact	Private	Protects customer communication details and contact records.
Case	Case	Private	Restricts service request visibility to authorized personnel only.
Utility Asset	Utility_Asset__c	Private	Secures infrastructure asset information and equipment identifiers.
Field Visit	Field_Visit__c	Private	Limits field service records to assigned operational users.

5.3.2 Criteria-Based Automated Sharing Rule

To support rapid response during critical operational events, an automated sharing rule is implemented within Salesforce.

Rule Configuration

Setting	Value
Rule Developer Name	High_Priority_Case_Sharing
Object Type	Case
Rule Type	Criteria-Based Sharing Rule
Criteria	Priority = High AND Status = New or Working
Shared With	Executive Management
Access Level	Read/Write

Business Purpose

This sharing rule automatically grants Executive Management access to high-priority service incidents. By bypassing standard visibility restrictions for critical cases, management personnel can monitor significant operational issues in real time and respond more effectively when escalation is required.

5.4 Salesforce Implementation and Configuration

This section outlines the configuration activities required to deploy the security framework within the Salesforce Developer Edition environment.

Step 1: Configure Organization-Wide Defaults (OWD)

The first stage of the implementation process involves securing baseline record visibility through Salesforce Sharing Settings.

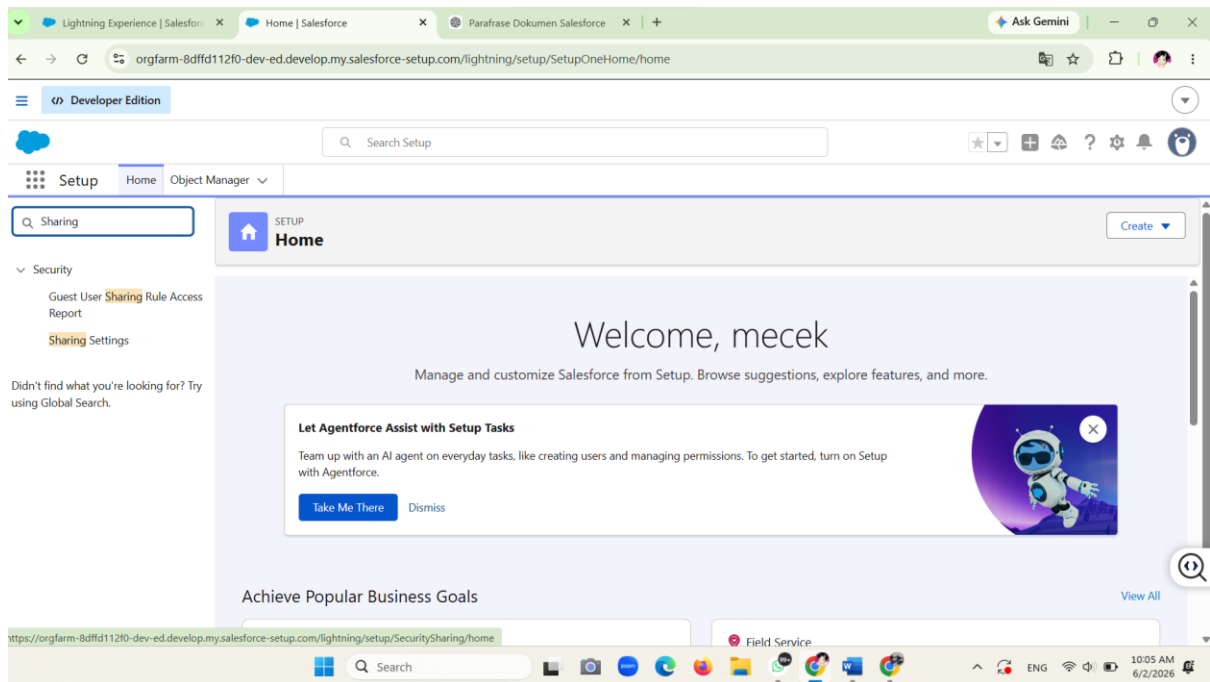
Navigate to:

Setup

→ Security

→ Sharing Settings

From this page, configure the Organization-Wide Default settings for all standard and custom objects according to the security model requirements defined in the previous sections.



- Click **Edit** on the Organization-Wide Defaults panel.
- Update the Default Internal Access column to **Private** for Account, Contact, Case, Utility_Asset__c, and Field_Visit__c.
- Click **Save & OK** and wait for the system data recalculation email.

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Ask Gemini

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Developer Edition

SetupHomeObject Manager

Sharing

Security

Guest UserSharing Rule Access Report

Sharing Settings

Didn't find what you're looking for? Try using Global Search.

SETUP

Sharing Settings

Help for this Page

This page displays your organization's sharing settings. These settings specify the level of access your users have to each others' data. Go to [Background Jobs](#) to monitor the progress of a change to an organization-wide default or a parallel sharing recalculation.

Manage sharing settings for: All Objects

Disable External Sharing Model

Default Sharing Settings

Organization-Wide Defaults

Object

Default Internal Access

Default External Access

Grant Access Using Hierarchies

Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Public Read/Write	Private	✓
Contact	Controlled by Parent	Controlled by Parent	✓
Order	Controlled by Parent	Controlled by Parent	✓
Asset	Controlled by Parent	Controlled by Parent	✓
Opportunity	Public Read/Write	Private	✓

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Ask Gemini

orgfarm-8dffd112f0-dev-ed.develop.my.salesforce-setup.com/lightning/setup/SecuritySharing/page?address=%2Fp%2Fshare%2FDefaultSharing%3Fid%3D00Dg50...

Developer Edition

SetupHomeObject Manager

Sharing

Security

Guest UserSharing Rule Access Report

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Didn't find what you're looking for? Try using Global Search.

SETUP

Sharing Settings

SaveCancel

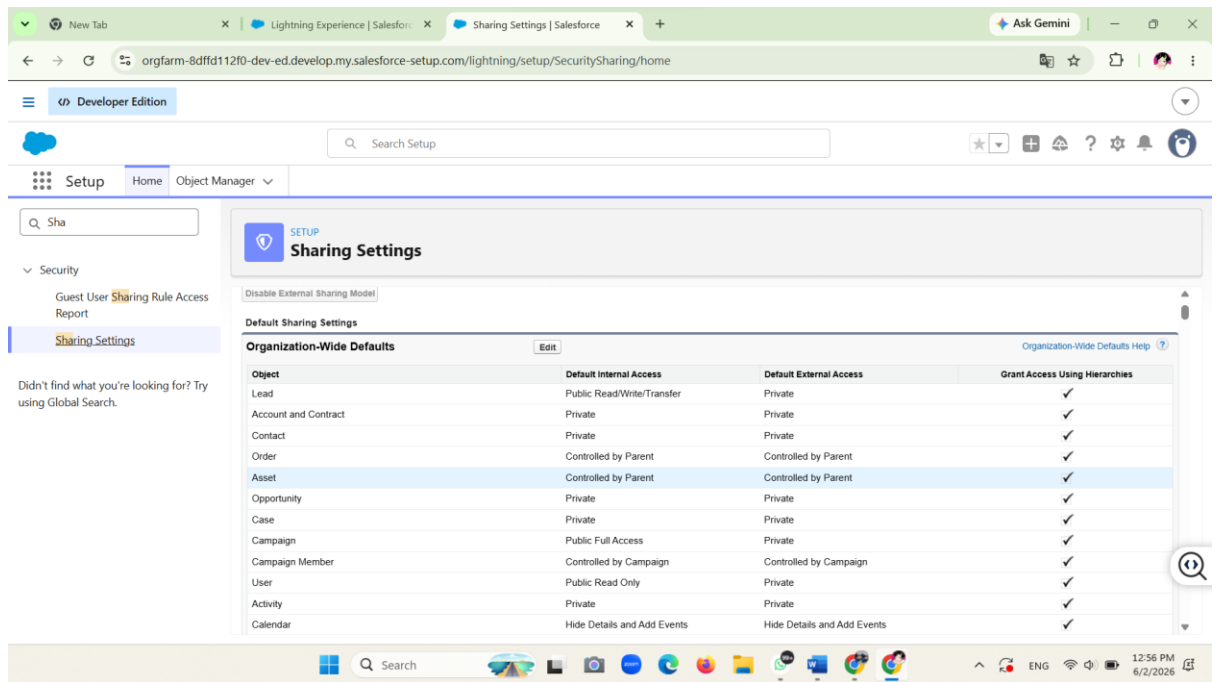
Object

Default Internal Access

Default External Access

Grant Access Using Hierarchies

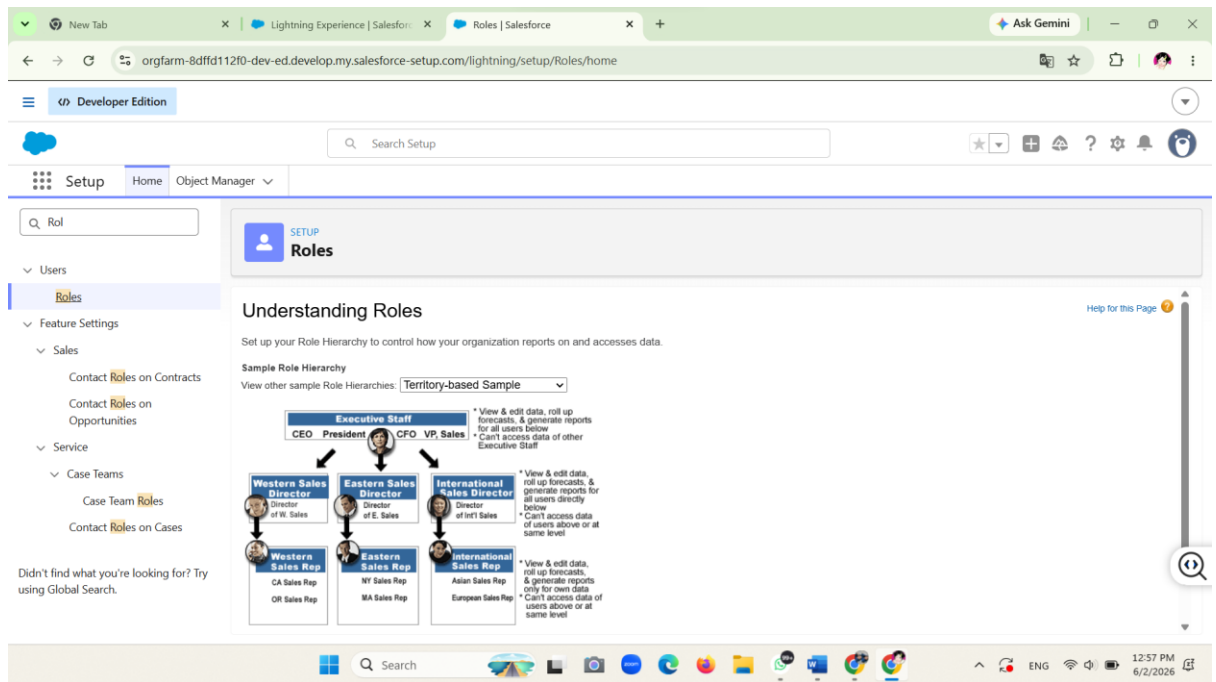
Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Private	Private	✓
Order	Controlled by Parent	Controlled by Parent	✓
Contact	Private	Controlled by Parent	✓
Asset	Controlled by Parent	Controlled by Parent	✓
Opportunity	Private	Private	✓
Case	Private	Private	✓
Campaign	Public Full Access	Private	✓
Campaign Member	Controlled by Campaign	Controlled by Campaign	✓
User	Public Read Only	Private	✓
Individual	Public Read/Write	Private	✓
Voice Call	Private	Private	✓
Activity	Private	Private	✓
Calendar	Hide Details and Add Events	Hide Details and Add Events	✓



Step 2: Role Hierarchy Structure Configuration

The top-down organizational tree was deployed to govern data visibility and automatic manager record roll-ups.

- Navigate to **Setup** → **Users** → **Roles**.
- Select the corporate hierarchy layout and click **Set Up Roles**.

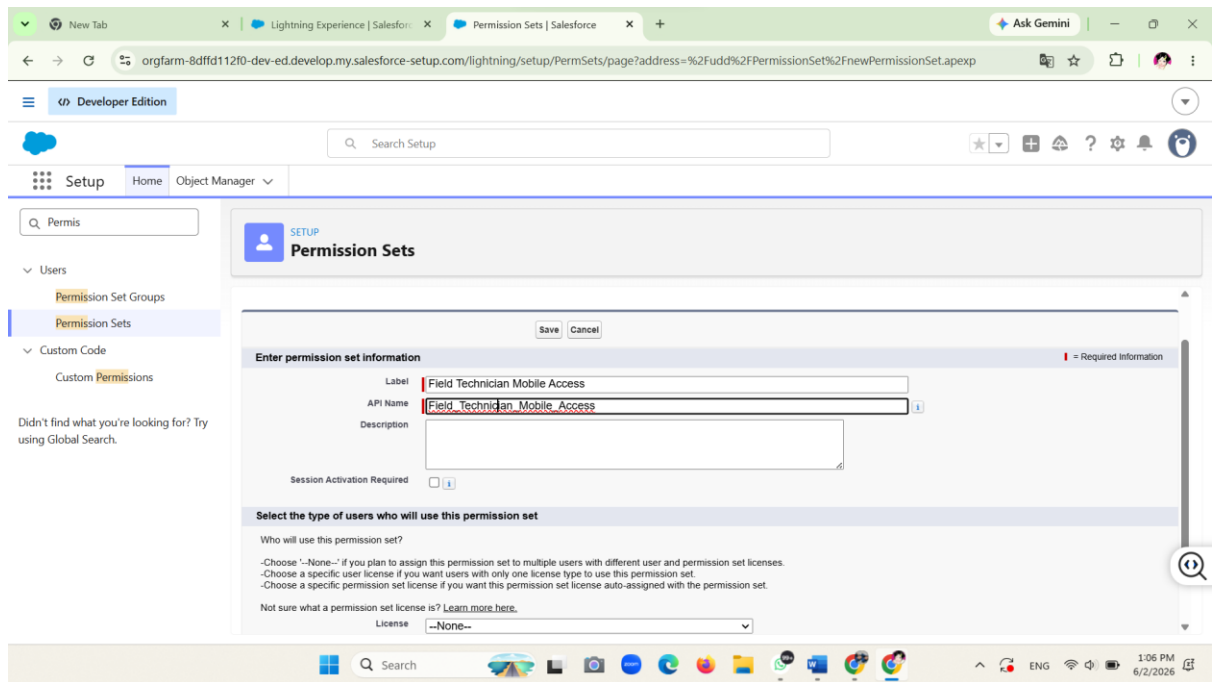


- c. Locate the default, out-of-the-box **CEO** role situated directly beneath the standard company root node, click **Edit**.
- d. Rename the Role Label to **Executive Management** (Role Name: **Executive_Managem**
- e. Click **Add Role** directly beneath the newly renamed **Executive Management** apex node to provision the first baseline child branch labeled **Customer Support Team**.
- f. Return to the **Executive Management** node, click **Add Role** again to provision the parallel baseline child branch labeled **Field Technicians**, and **Save** it.

Step 3: Permission Set Provisioning for Field Technicians

A specialized permission container was built to expand object-level and field-level rights for remote operations without altering base system profiles.

- a. Navigate to **Setup** → **Users** → **Permission Sets** and click **New**.
- b. Label the record **Field Technician Mobile Access** and click **Save**

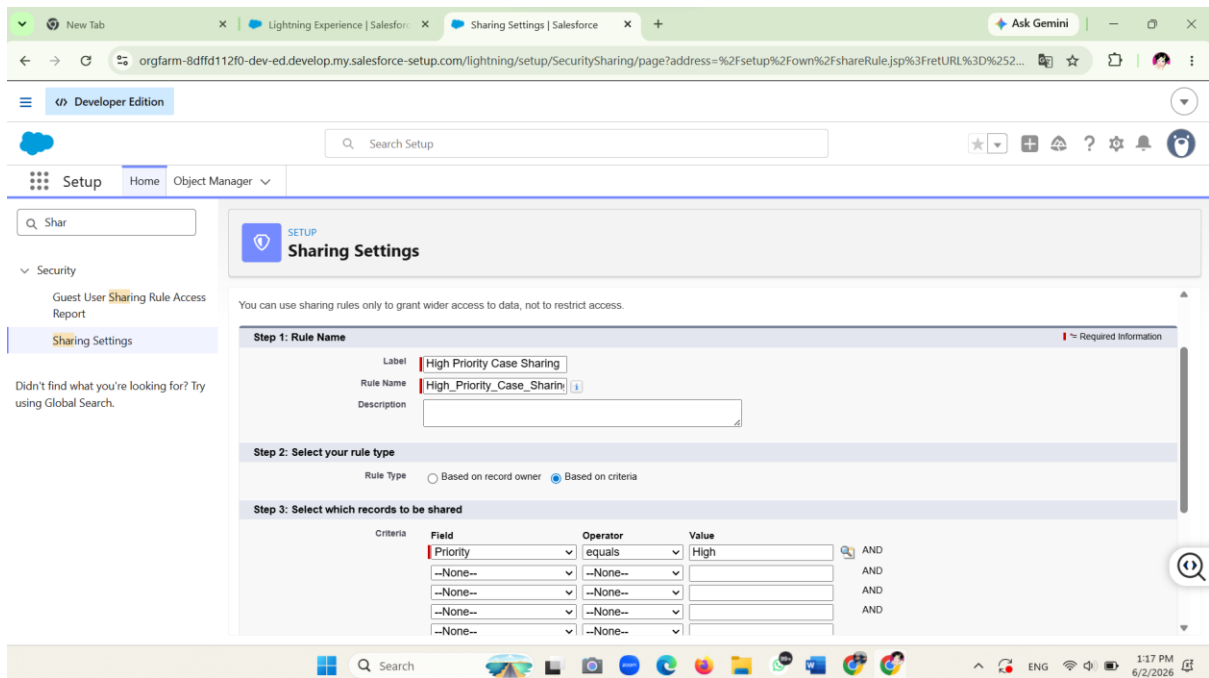


- c. Open **Object Settings** → scroll down and select **Field Visits** → click **Edit**
- d. Enable **Read, Create, and Edit** checkboxes. Under Field Permissions, check **Edit Access** for **Repair Status** and **Technician Notes**.
- e. Click **Save**.

Step 4: Automated Case Sharing Rule Deployment

The system data sharing exception was activated to pass high-priority case rows to management queues automatically.

- a. Scroll down the **Sharing Settings** home layout to the **Case Sharing Rules** related list and click **New**.
- b. Label the rule **High Priority Case Sharing** → Set Rule Type to **Based on criteria**.
- c. Set Criteria to: **Priority EQUALS High**.
- d. Set Share With to: **Role** → **Executive Management**.
- e. Set Access Level to: **Read/Write** for Case Access.

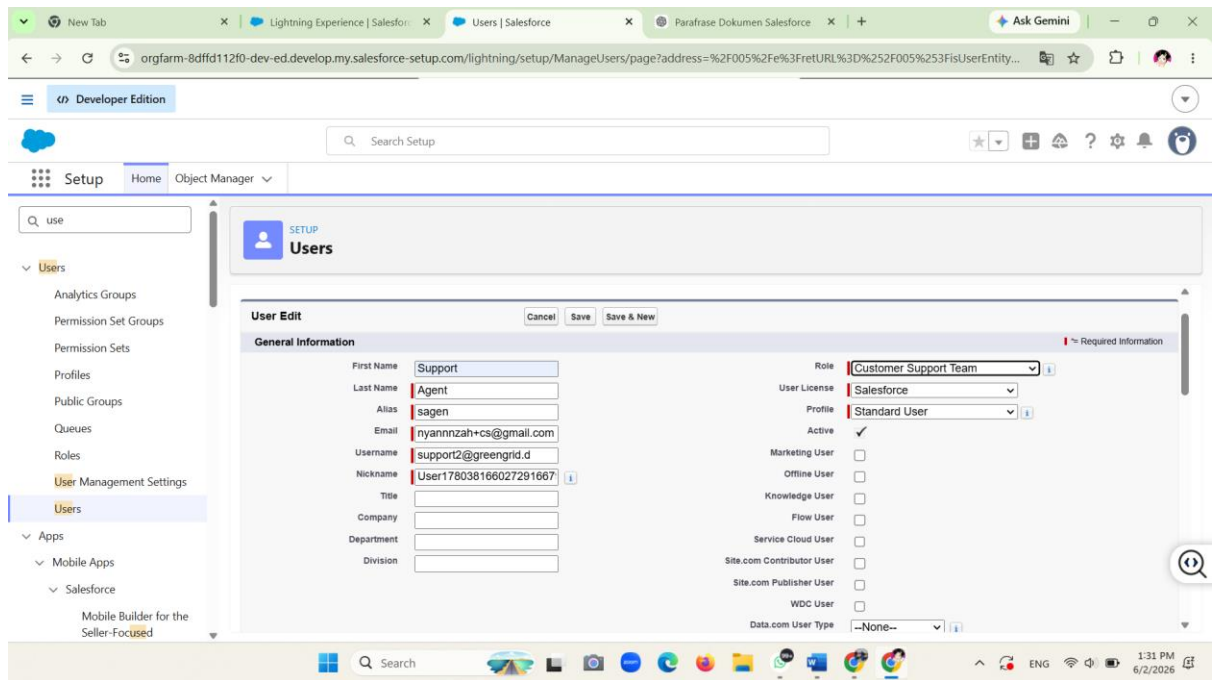


f. Click **Save** (click **OK** if pop up appears)

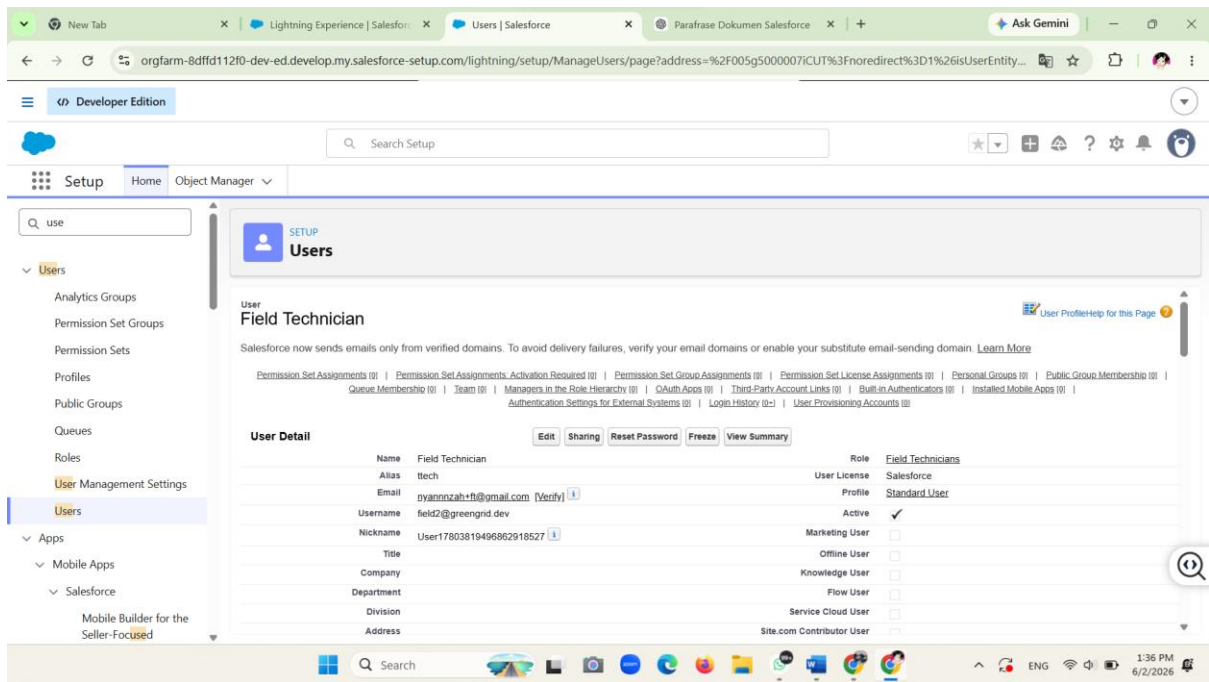
Step 5: Operational User Provisioning and Role Assignment

To validate that the Private Organization-Wide Defaults and vertical Role Hierarchy function correctly under live operational parameters, standard test user identities were provisioned.

- Navigate to **Setup** → **Users** → **Users** → click the **New User** button.
- Provision the first test user representing the support desk environment:
 - First Name:** Support | **Last Name:** Agent
 - Email / Username:** Use a distinct corporate string format (e.g., support_desk@greengrid.dev).
 - Role:** Assign exactly to the **Customer Support Team** node.
 - User License:** Select **Salesforce**.
 - Profile:** Select the **Standard User** profile.



- c. Click **Save & New**.
- d. Provision the second test user representing the remote engineering mobile workforce:
 - o **First Name:** Field | **Last Name:** Technician
 - o **Email / Username:** Use a distinct unique string format (e.g., `field_crew@greengrid.dev`).
 - o **Role:** Assign exactly to the **Field Technicians** node.
 - o **User License:** Select **Salesforce**.
 - o **Profile:** Select the **Standard User** profile.
- e. Click **Save**



- f. Return to the primary **Active Users** list view directory page to verify that both the **Support Agent** and **Field Technician** record entries successfully display active markers, matching security profile parameters, and accurate vertical role alignments.

